AI FOR GOOD · HACKATHON 6 “LIVE PULSE” · SDG 9

Starlink Pulse

A live sentiment tracker for satellite-internet experience, for a rural municipality deciding on a Starlink pilot or subsidy

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Bluesky stream · Hugging Face transformer · Streamlit dashboard

THE PROBLEM

Coverage maps say where internet exists, not whether people can rely on it

90.5%

of EU rural households had internet access in 2023, against 94.9% in cities

Eurostat, Urban-rural Europe: digital society

Satellite fills the gap

Rural areas without fibre increasingly depend on Starlink. Councils are asked to recommend or subsidise it.

Nobody reads along

Users report outages and slow speeds online within minutes, but no one can read 6,000+ posts a day.

SDG 9, target 9.c: “universal and affordable access to the Internet”. Reliable infrastructure, not just available infrastructure.

THE USER AND THE DECISION

A rural municipality’s broadband officer

The decision

Should we run a **Starlink pilot** (a small trial in a few homes, schools or farms) or **subsidise** it for households with no fibre or good mobile signal?

What they do with the pulse

Watch for rising complaints, open the posts behind a change, and decide what to check with real network data before advising councillors.

Whose voices are missing

People who don’t use Bluesky, don’t write in English, don’t post publicly, or are offline. Someone whose connection is down can’t post about it.

THE DATA STREAM AND ITS TERMS

Bluesky posts that mention Starlink

Why not Reddit?

We first chose r/Starlink, but Reddit's Responsible Builder Policy requires explicit approval before any API access. None arrived before the deadline.

Bluesky

Open public search API, free account and app password, English posts only. We only read: no posting or liking.

What we store, and nothing else

Post text (links removed, @names → @user), times, the model's label and scores. **No names, handles, profiles or links.**

HOW THE TOOL WORKS

Six steps, every 30 minutes

1
Schedule
GitHub Actions starts the collector on its own servers

`collect.yml`

2
Fetch
Logs in to Bluesky and fetches new English posts

`collector.py`

3
Model
The transformer scores every new post

`collector.py`

4
Filter
Keeps posts about the connection, not politics

`scope.py`

5
Store
Adds text, time and scores to a CSV on GitHub

`posts.csv`

6
Dashboard
Streamlit reads the CSV and draws the charts

`app.py`

No duplicates, no gaps
Each post gets a short hash of its address, so a run skips posts it already has and catches up on everything since the last run.

Runs in the cloud
Free on GitHub Actions; a laptop can be off. The Bluesky app password is a GitHub secret, never in the code.

WHAT THE MODEL DOES

How the transformer reads a post

1 · Split into tokens

The text is cut into word pieces and turned into numbers.

2 · Read in context

RoBERTa weighs every word against all the others, so “offline… couldn’t download” is read as one complaint.

3 · Three probabilities

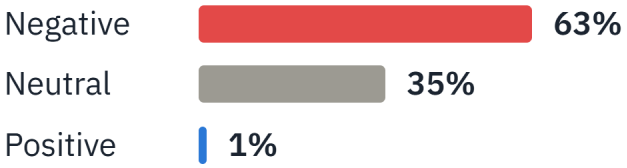
Negative, neutral and positive, adding up to 100%.

4 · Pick the label

The highest wins. We store all three, so the dashboard can show how sure it was.

A REAL POST FROM OUR STREAM

“Starlink was offline so I couldn’t download my ticket”



Label: negative

cardiffnlp/twitter-roberta-base-sentiment-latest · trained on ~124M tweets (2018–2021), fine-tuned on TweetEval · English · CC BY 4.0

Why this one: short social posts like ours, a separate neutral label, and free to run on a normal CPU, so no paid API.

THE TOPIC FILTER

Most Starlink talk is politics

86%

of the posts our first filter kept (925 of 1,071) were about Ukraine, Musk, US politics or rockets

Rule 1 (first version)

Mentions the connection: internet, outage, speed, dish, rural, service…

Rule 2 (added after reading our data)

And none of: Ukraine, Russia, Musk, Trump, war, drone, SpaceX, rocket, launch…

Result: 1,071 → about 155 posts

Now real experiences: “I need stable service at my remote cottage to monitor the thermostat & sump pump”

Trade-off: a word list is blunt

“Down again, thanks Elon” is dropped because of “Elon”; war posts with “putler” or “Moscow” still slip through.

WHAT IT COLLECTED

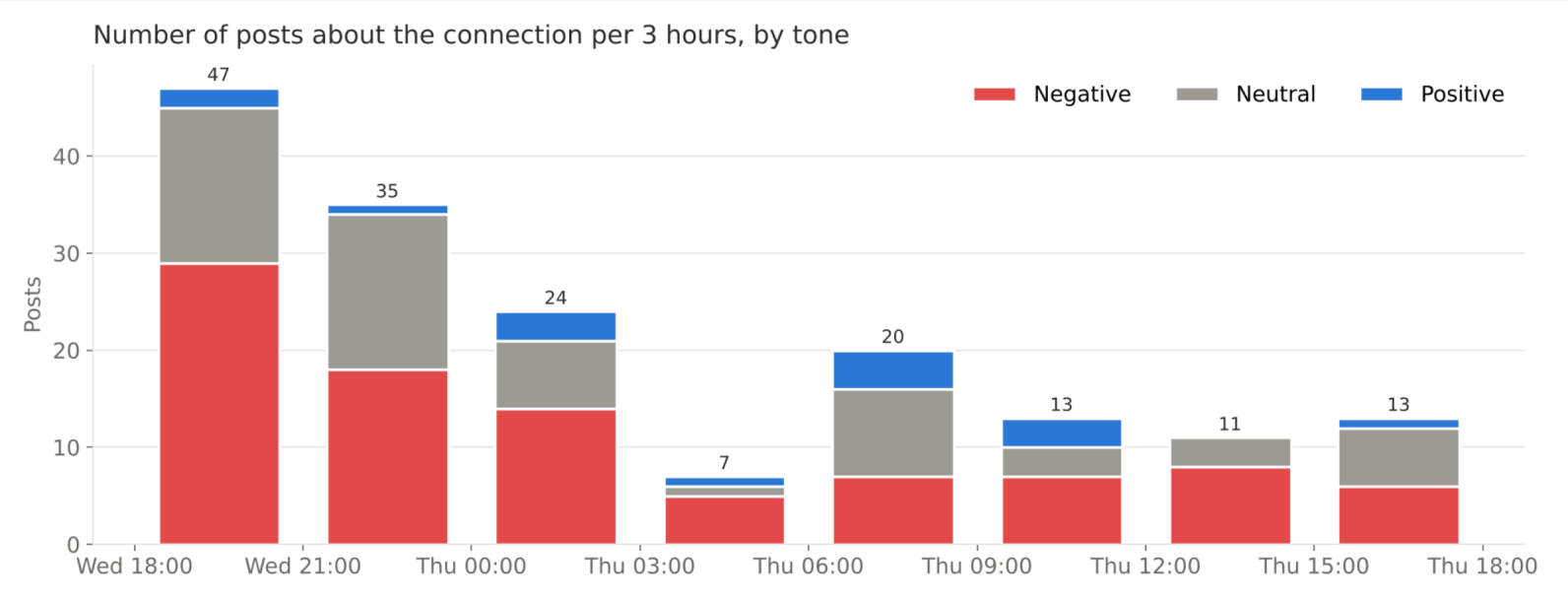
6,542 posts, 170 about the connection

6,542
posts scored by the model, 14 Oct 18:27 → 15 Oct 17:42

1,012
collected live (4 runs) + 5,530 in the first catch-up run

170
about the connection after the filter

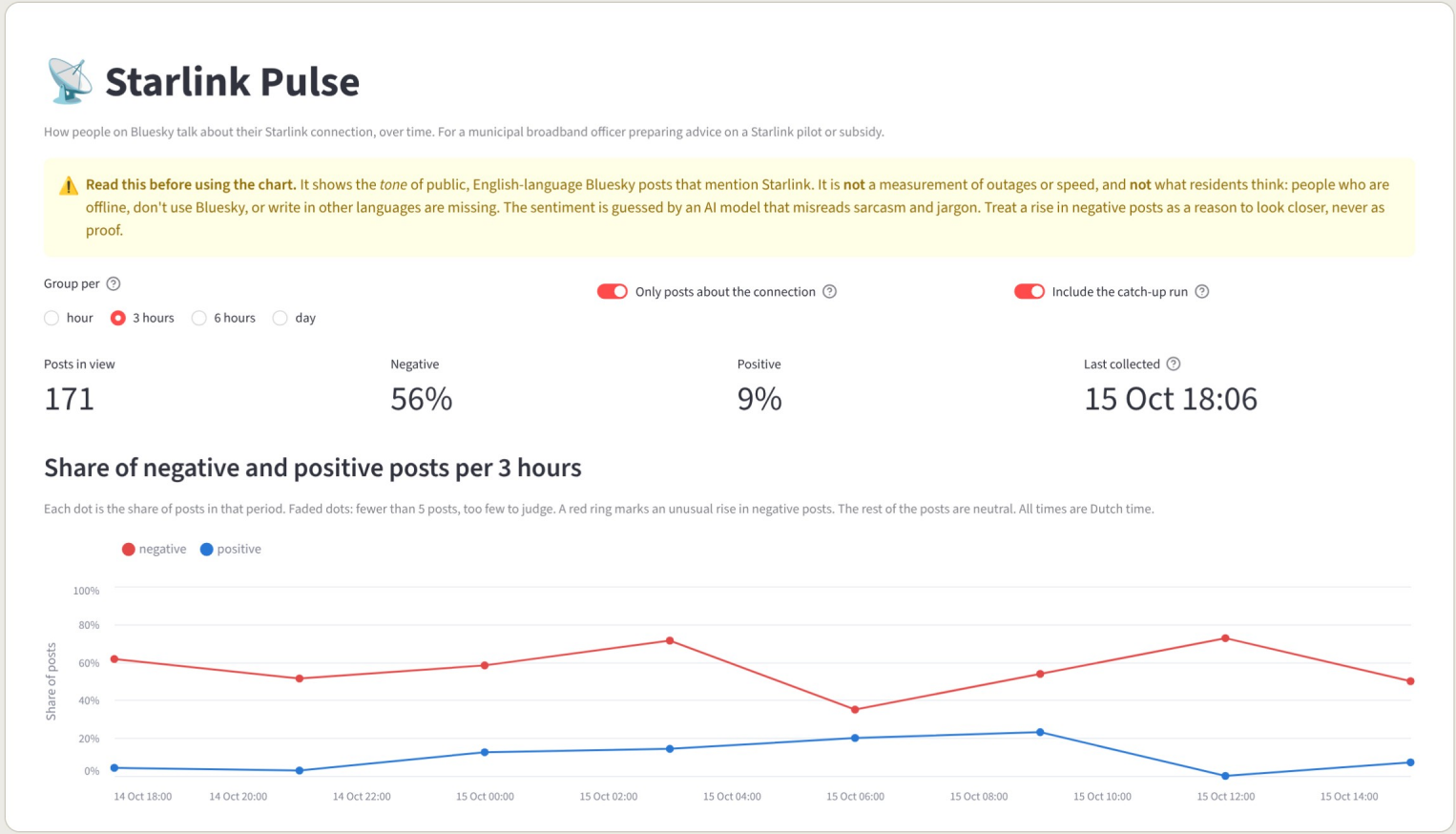
55% · 36% · 9%
negative · neutral · positive



Numbers as of 15 Oct 2026, 17:43 Dutch time. The collector keeps adding posts after the deadline.

THE LIVE DASHBOARD

starlink-pulse.streamlit.app



Warning box

Tone of posts, not outages or public opinion

Controls

Hour / 3 h / 6 h / day, topic filter on or off, catch-up run in or out

Headline numbers

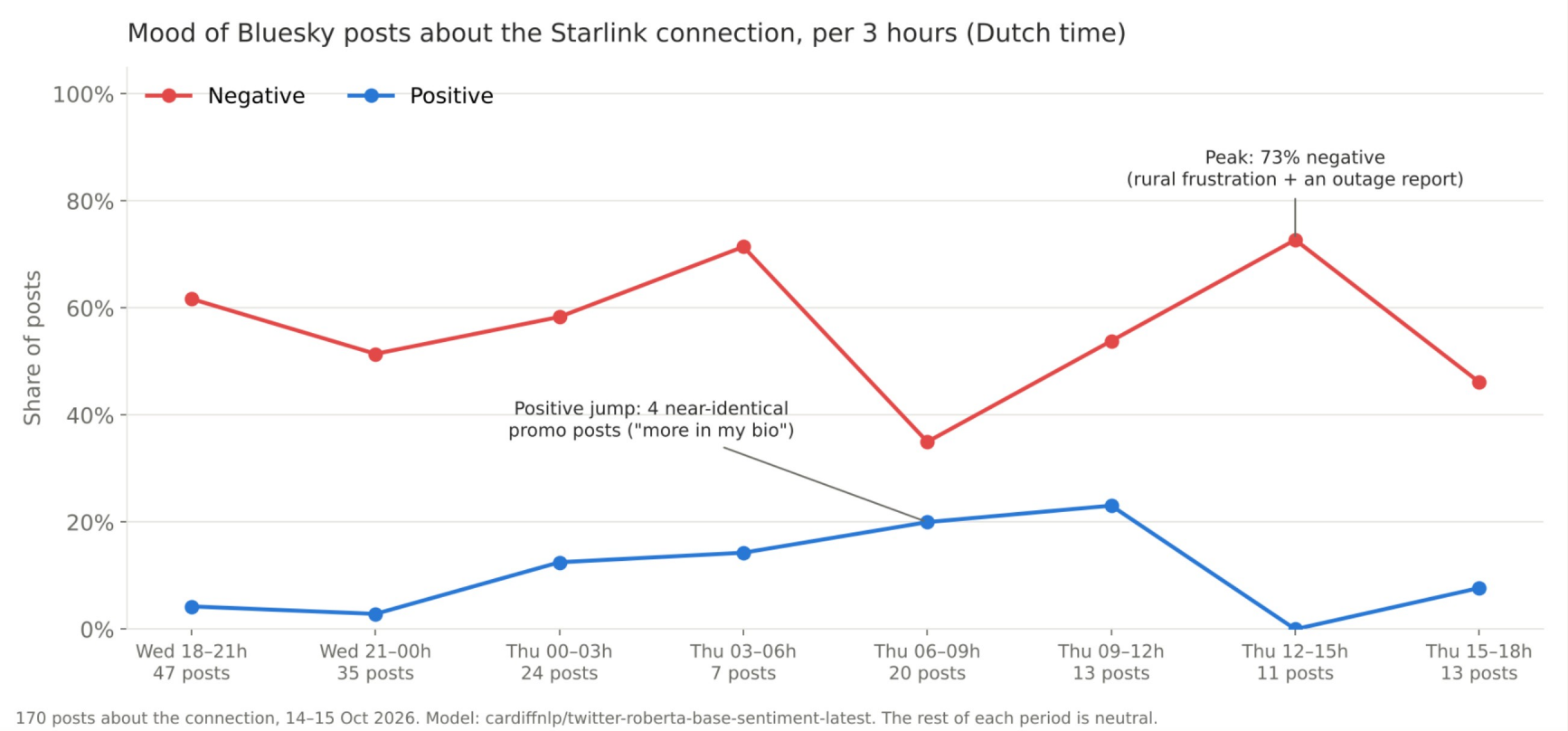
Posts in view next to every percentage

Mood over time

Red = negative, blue = positive; thin periods faded

THE MOOD OVER TIME

Over half of these posts are negative



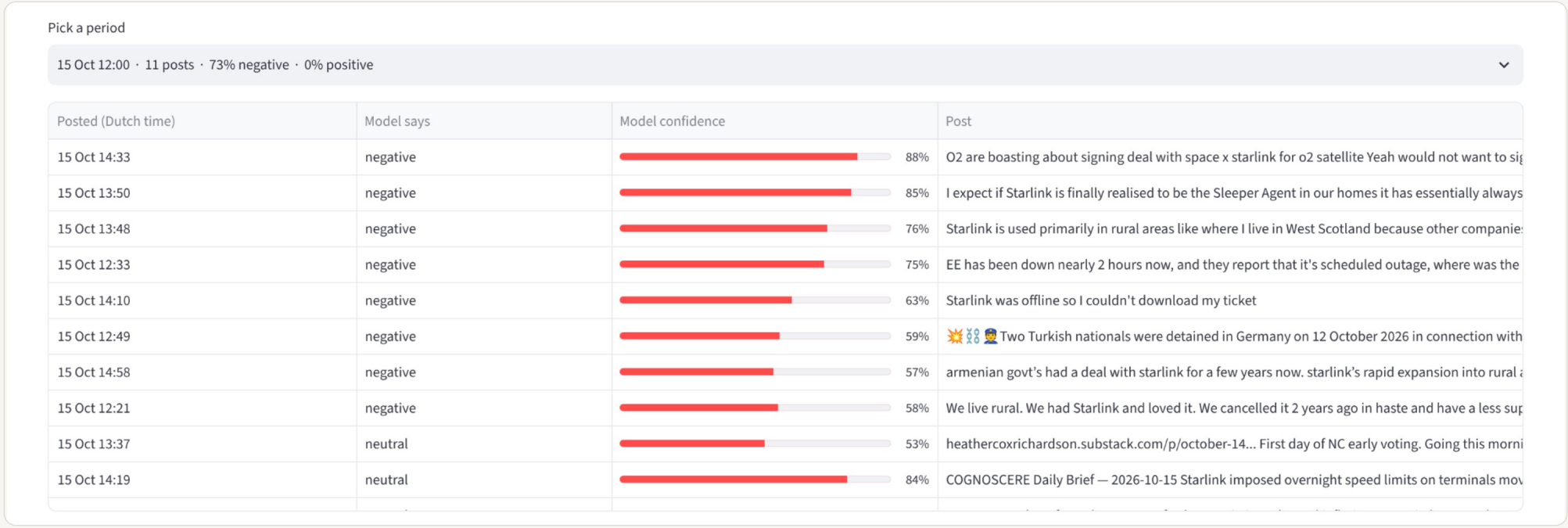
170 posts about the connection, 14–15 Oct 2026. Model: cardiffnlp/twitter-roberta-base-sentiment-latest. The rest of each period is neutral.

Negative stays between 35% and 73%; positive rarely tops 20%.

Two moments stand out. Next: what is behind them.

BEHIND THE PEAK · THU 12:00–15:00 · 73% NEGATIVE

Rural frustration and one outage report



Posted (Dutch time)	Model says	Model confidence	Post
15 Oct 14:33	negative	88%	O2 are boasting about signing deal with space x starlink for o2 satellite Yeah would not want to sig
15 Oct 13:50	negative	85%	I expect if Starlink is finally realised to be the Sleeper Agent in our homes it has essentially always
15 Oct 13:48	negative	76%	Starlink is used primarily in rural areas like where I live in West Scotland because other companie
15 Oct 12:33	negative	75%	EE has been down nearly 2 hours now, and they report that it's scheduled outage, where was the
15 Oct 14:10	negative	63%	Starlink was offline so I couldn't download my ticket
15 Oct 12:49	negative	59%	💥⛓️👮 Two Turkish nationals were detained in Germany on 12 October 2026 in connection with
15 Oct 14:58	negative	57%	armenian govt's had a deal with starlink for a few years now. starlink's rapid expansion into rural a
15 Oct 12:21	negative	58%	We live rural. We had Starlink and loved it. We cancelled it 2 years ago in haste and have a less sup
15 Oct 13:37	neutral	53%	heathercoxrichardson.substack.com/p/october-14... First day of NC early voting. Going this morni
15 Oct 14:19	neutral	84%	COGNOSCERE Daily Brief — 2026-10-15 Starlink imposed overnight speed limits on terminals mov

Reading it: "Starlink was offline so I couldn't download my ticket" is a real outage report; "other companies left us all high and dry" is rural dependency. Two posts (an arson attack, a politics newsletter) slipped past the filter. For the officer: a reason to look at rural dependency, not proof of an outage.

BEHIND THE DIP · THU 06:00–09:00 · 20% POSITIVE

Four promo posts moved the line

Posted within two minutes, same template, all labelled positive:

> *“Starlink plays an important role in connectivity. Could this speed cap change things on the ground? More context via my bio.” (posted twice)*

> *“Starlink is more than just satellite internet in this situation… More context in my bio.”*

> *“Technology and geopolitics are becoming increasingly connected… More in my bio.”*

Why it matters

The brief asks: “Can bots or one very active account move your line?” In our data, yes. Without these four, this period looks like the rest.

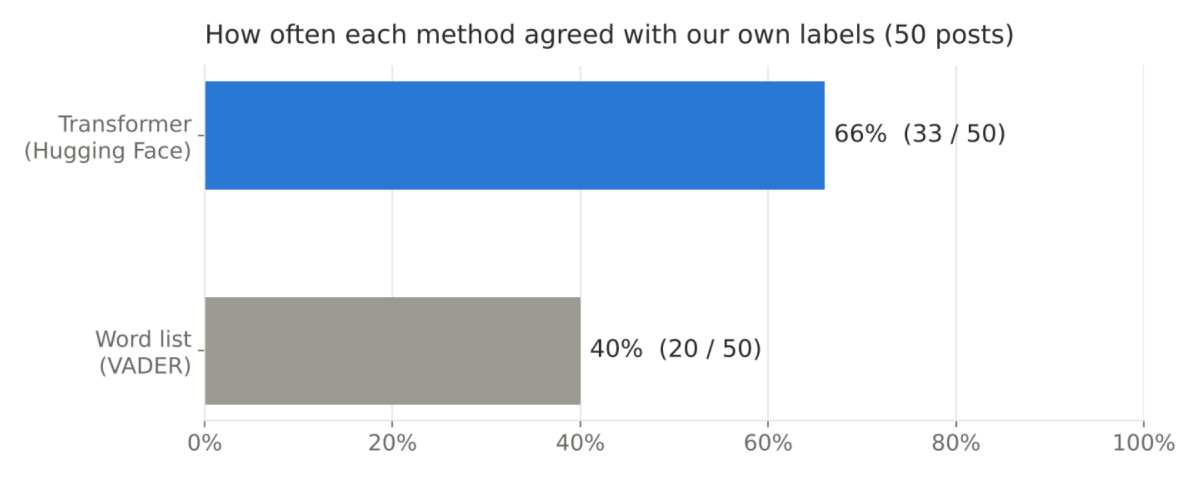
We store no usernames, so we can’t prove it was one account, but the pattern is clear.

What protects the user: post counts beside every point, and the posts behind it one click away.

THE HONEST TEST

50 posts we labelled ourselves

50 random posts about the connection, labelled negative / neutral / positive by us **before** seeing any model output.



Transformer's mistakes

Us ↓ / model →	neg	neu	pos
negative (26)	19	7	0
neutral (18)	6	10	2
positive (6)	1	1	4

Its errors are almost all **negative ↔ neutral**. It never called one of our complaints positive. VADER did that 9 times.

THE COMPARISON

Counting words vs reading the sentence

VADER (word-list baseline)

A fixed list of about 7,500 words with scores (“great” +, “terrible” –). It adds them up, with simple rules for “not”, CAPITALS and “!”. Result –1 to +1; above +0.05 = positive. Transparent, but it never learns and does not understand context.

Transformer (our model)

Learned from about 124 million tweets what positive, neutral and negative look like, and reads every word in relation to the rest of the post. Smarter, but a black box: it can’t tell you why.

“United Airlines adds Starlink Wi-Fi - Thus making sure that I’ll never fly with them ever again. Thanks for making the choice so easy UA!”

Us: **negative** VADER: **positive (+0.90)**, “thanks”, “easy” Transformer: **negative (74%)**

WHERE THE TRANSFORMER GOES WRONG

Three failures that taught us the most

A joke with a rude word

"...internet here is piss poor... 9 of them have starlink attached to their roof. Guess which roof doesn't have starlink."

Us: **neutral** · Model: **negative (87%)**

"Piss poor" dominated; the writer is joking, not complaining about Starlink.

A question that sounds negative

"Are starlink speeds actually measured in km/h? That's a pretty weird unit..."

Us: **neutral** · Model: **negative (68%)**

A genuine question, but "weird" pulled it to negative.

Negative without negative words

"Time for all former US alies to move away from US internet infrastructures, starting with Starlink."

Us: **negative** · Model: **neutral (59%)**

A call to boycott with no single negative word. VADER missed it too.

Also: some war posts still slipped through our topic filter ("putler", "Moscow", "interceptors"), because those exact words weren't on our list.

WHICH ONE WOULD WE HAND TO THE OFFICER?

The transformer, with a warning

Why

66% vs 40% agreement with us, and it never mistook a complaint for praise, the error that would mislead an officer most.

But not on its own

About one in three posts is labelled differently from how we read it. So: counts beside every score, and read the posts behind a peak before acting.

ETHICAL REFLECTION

Risks of this stream and this model

The risks

- **Missing voices:** offline, non-English and non-Bluesky users; people with a broken connection can't post.
- **Loud few:** one account or a bot can move a period (our promo posts).
- **Misreading:** 1 in 3 labels differ from ours; jokes and implied criticism trip it.
- **Over-trust:** a council treating the line as an outage measure could make a poor subsidy decision.
- **Privacy:** these are real people's words.

What we did

- No names, handles, profiles or post links stored; @mentions replaced.
- Post count beside every score; periods under 5 posts faded.
- Warning on the dashboard: tone, not outages or opinion.
- Every peak or dip opens to the posts behind it.
- Tightened the topic filter after reading our own data.
- Published our 50-post test, mistakes included.

Still open: Bluesky's developer guidelines ask services to delete content a user deletes. Our tool doesn't re-check that yet.

HONEST LIMITS AND NEXT STEPS

What didn’t go to plan

48 hours of running

The tool first ran on 15 October. A one-off catch-up run collected the posts of the day before; every post keeps its real time, and the dashboard can hide the catch-up posts.

GitHub’s 30-minute schedule

GitHub calls scheduled runs “best effort”. On our new repo it took hours to start, so some runs were started by hand. Each run catches up on everything since the last.

Next steps

- An outside trigger (e.g. cron-job.org) that starts the collector exactly every 30 minutes
- Drop near-identical posts automatically, against bots and promo accounts
- Remove posts their authors delete, as Bluesky’s guidelines ask
- A multilingual model for Dutch posts, and fine-tuning on posts we label ourselves

TRY IT YOURSELF

Starlink Pulse is still running

Live dashboard

starlink-pulse.streamlit.app

Code, data, README and test

github.com/cameronruiter/starlink-pulse

Model

cardiffnlp/twitter-roberta-base-sentiment-latest

It must not be used for

- Measuring outages, speed or uptime
- Claiming what residents or users think
- Deciding a subsidy without network data or local research
- Identifying or judging individual posters
- Marketing for or against Starlink